ARJUNA

v2 · Adaptive Risk & Regime Allocation System

AQR / Bridgewater-style risk investing — for retail Indian equities

2.04	12.8%	0.996	0.00
Sharpe	max DD	Defl. Sharpe	PBO

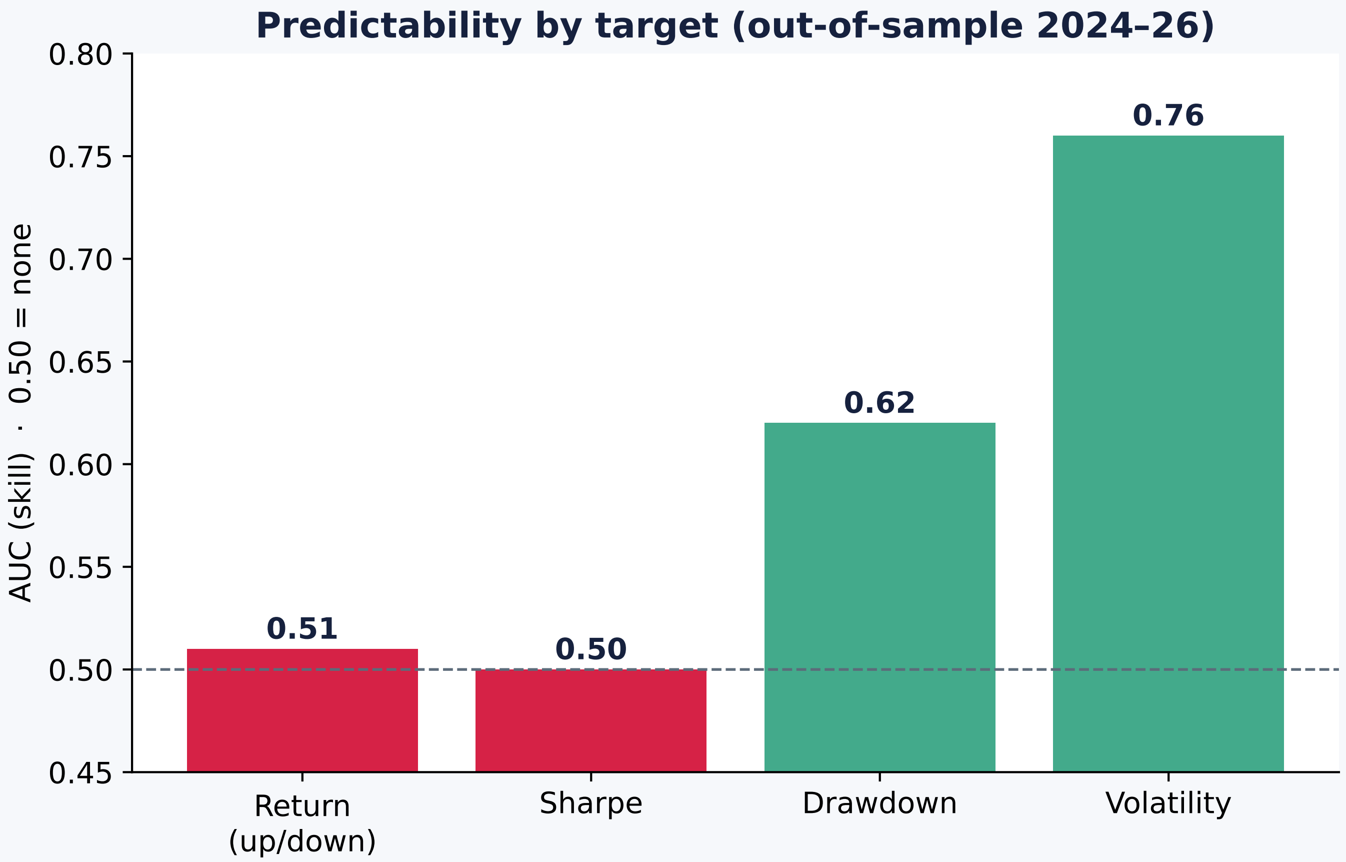
NOT a stock-picking bot. A risk-allocation & regime-management bot.

Validated on clean Angel One data · Nifty-200 · net of cost
Returns are unpredictable. Risk is. ARJUNA forecasts risk, not winners.

THE CORE PRINCIPLE

Returns are noise. Risk has structure.

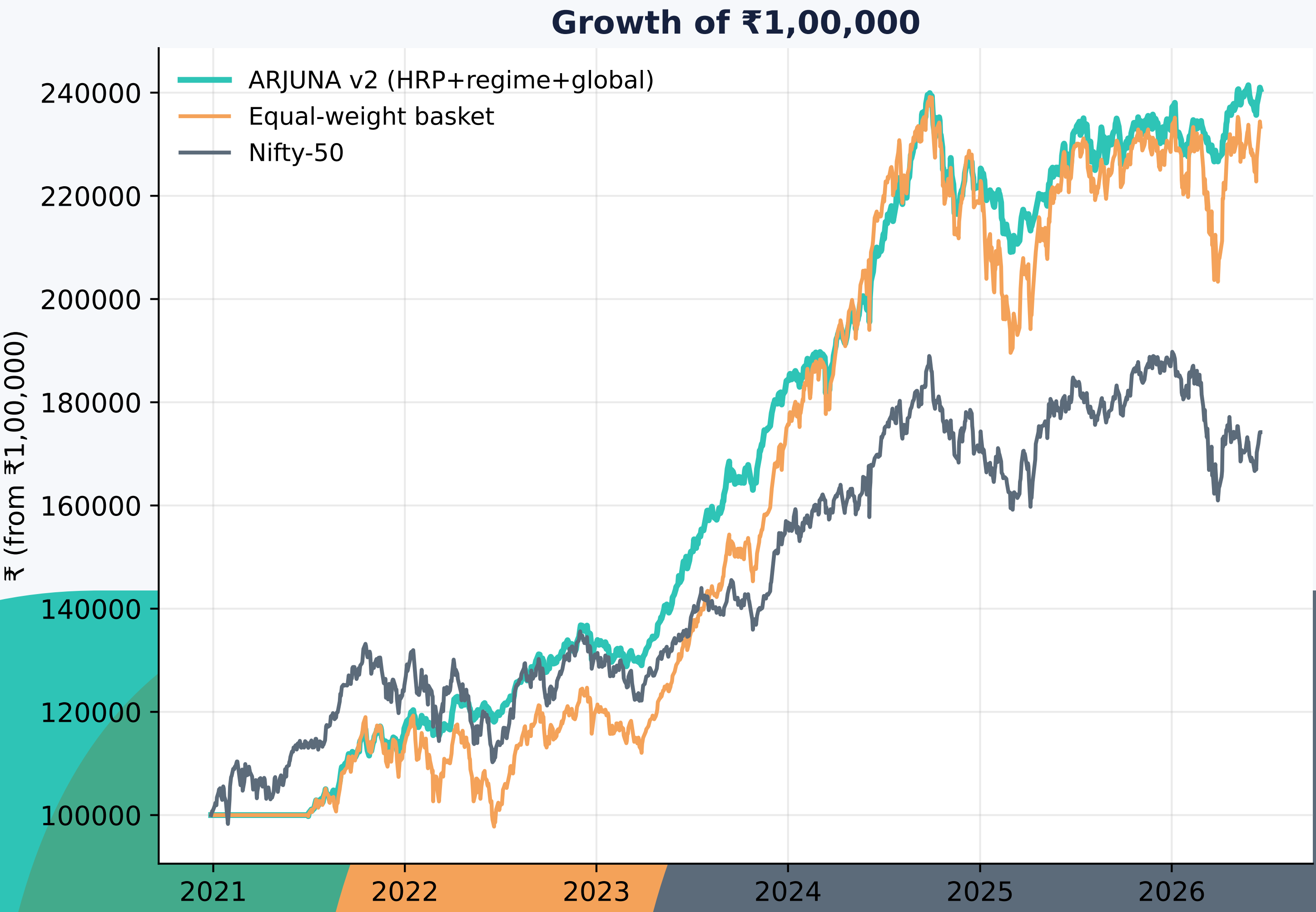
We tested 13 AI model families on 'will the stock go up?' Every one scored ~0.50 (a coin flip). Then we changed the TARGET — and the same model suddenly had real skill at predicting RISK.



So ARJUNA forecasts RISK · REGIME · EXPOSURE · WEIGHTS — never returns.
Objective: long-term risk-adjusted compounding. Survival first; raw CAGR last.

VALIDATED RESULTS

Champion vs the index (₹1L, ~5.5y, net of cost)



17.7%	**2.04**	**12.8%**	**0.80**
CAGR	Sharpe		

Rigour gate (3 independent checks):

Deflated Sharpe 0.996 ✓	PBO 0.00 ✓	Purged walk-fwd ✓

Survivorship inflates absolute CAGR → trust the Sharpe/dra

ARCHITECTURE

Four layers — risk in, portfolio out

LAYER 1 · MARKET STATE

Global Risk Engine (S&P/VIX/USD) · regime (VIX+200DMA) · breadth · FII/DII

↓

LAYER 2 · RISK FORECAST

Volatility (AUC 0.76) · drawdown · trailing covariance

↓

LAYER 3 · CORRELATION ENGINE

Ledoit-Wolf covariance · HRP clusters · sector caps (20 stocks ≠ 20 bets)

↓

LAYER 4 · PORTFOLIO

HRP / min-var / inverse-vol weights · per-name cap · news blow-up filter

FINAL PORTFOLIO
(monthly rebalance · paper→live)

AI is used for risk, regime & news — NOT for 'which stock doubles'.

THE DISCIPLINE

Evidence killed hypotheses — only survivors ship

✓ SURVIVED (risk & regime)	✗ REJECTED (tested, no edge)
Global Risk Engine	13 ML return-models (~0.50)
Regime (VIX + 200-DMA)	HMM regime (1.06 < 1.64)
Breadth · FII/DII (live)	RL · GNN · Transformers
HRP / min-var / inverse-vol	GARCH (no gain vs trailing)
Volatility forecast (0.76)	Vol-targeting (just levers up)
News blow-up filter	Crash classifier (0.56, wash)
Broad diversification	Concentration / multibaggers

Every rejection made ARJUNA simpler and more honest. Data is the bottleneck — not models.

HOW TO USE IT

From config to compounding

1 · CONFIGURE	india/config.py — universe, method=hrp, regime=global, capital. One file, no logic edits.
2 · DAILY RUN (auto)	Scheduled task 'ArjunaDailyPaper' (weekdays 18:00): pull data → FII/DII → news → portfolio → paper log.
3 · GET THE BASKET	python india/run_arjuna.py --capital 100000 → risk-weighted holdings, regime-scaled, news-filtered.
4 · SIZE TO CAPITAL	Dynamic allocator auto-fits ₹5K→₹5L (more capital = more names; ₹25K+ deploys fully).
5 · PAPER FIRST	Forward paper log (output/paper_log.csv) for weeks–months → unbiased verdict, no survivorship.
6 · GO LIVE	Only after paper confirms: rotate Angel credentials, fund, wire order placement.

Golden rule: keep cash out until the FORWARD paper run beats the index, net of cost.

HONEST EXPECTATIONS

What it will & won't do

WILL: beat the index on risk-adjusted terms · cut drawdowns · de-risk in global/regime stress · diversify by correlation · avoid news blow-ups · compound steadily.

WON'T: pick the next BSE/multibagger (impossible in advance — proven) · predict returns · shoot the lights out every year (it's streaky: great in trends, flat in chop).

The real ceiling = DATA (point-in-time India fundamentals), not models.

Roadmap (free-data, Lab → Core only via DSR>0.95 + PBO):

- Fama-French factor overlay
- Conformal uncertainty bands
- Triple-barrier + meta-labelling
- Point-in-time fundamentals (the unlock)

ARJUNA Doctrine: survival > prediction · diversification > concentration · regime > selection · construction > models · data > complexity · robustness > backtests.

ARJUNA v2 — built on evidence, not hope.